

Predicting **NBA Draft** from NCAA Player Statistics

INFO 5368-030 Final Project

Xinda Ding, Je-Wei Tsao, Pin-Hsuan Lai, Yui To Wong, Pin-Yeh Lai, Changyi Zhou

1. Introduction

Motivation & Problem Statement

The Problem

Drafting is a high-stakes selection problem. A missed pick consumes salary-cap resources, roster slots, and developmental attention, while a successful pick creates long-term franchise value.

However, college statistics are heavily obscured by conference strength, team role, pace, and uneven opposition quality.

The Objective

Can a transparent machine learning pipeline built **only from public NCAA player statistics** estimate NBA draft outcomes well enough to support front-office screening?

We aim to build a reproducible decision-support tool that ranks players and highlights key signals.

Technical Focus & Impact



Methodology

Implementation of Logistic Regression, Multilayer Perceptron (MLP), and K-Nearest Neighbors (KNN) using NumPy and Pandas.



Novelty

Operating entirely on accessible, public data rather than proprietary tracking systems, ensuring total reproducibility and democratization of scouting analytics.



Social Impact

Reduces subjective human bias in talent evaluation. However, it raises ethical concerns regarding reducing athletes to pure numbers and missing intangible human contexts.

2. Background & Prior Work

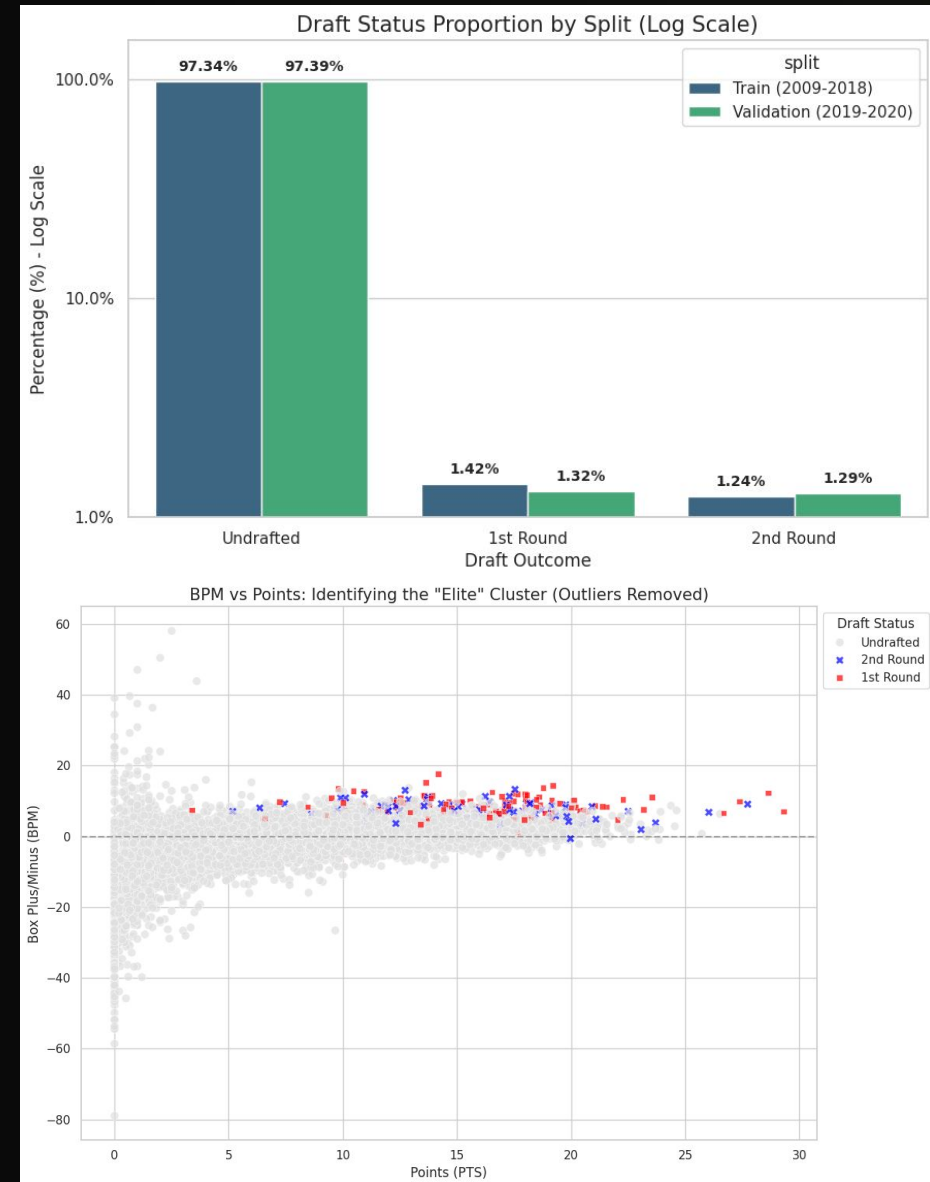
|| Prior sports analytics work shows predictive signal, but often relies on proprietary tracking feeds or subjective draft order. We focus strictly on draft status using reproducible public data. ||

— Bridging Accessibility and Predictive Ambition

While Berri (2011) focused on draft order and Patton (2021) utilized proprietary spatial tracking, our pipeline tests if standard box-score and engineered metrics can project categorical draft success.

3.1 Data Collection & Exploration

-  **Source:** Kaggle "College Basketball 2009-2021 + NBA Advanced Stats".
-  **Size:** Over 25,445 player-season records spanning 67 features.
-  **Target Labels:** 0 (Undrafted), 1 (2nd Round), 2 (1st Round).
-  **Extreme Imbalance:** ~97.3% of the training dataset consists of undrafted players, requiring weighted losses.
-  **EDA Findings:** Elite cluster analysis revealed heavy overlap between high-performing undrafted players and drafted prospects.



Data Processing Pipeline

-  **Cleaning & Imputation:** Conducted schema cleanup, season-level missingness audits, and median imputation for numerical features. Handled categorical missingness via an 'Unknown' bucket.
-  **Encoding:** Applied one-hot encoding for categorical variables including team, conference, and player role.
-  **Normalization:** Performed Z-score scaling for numerical inputs using statistics derived strictly from the training set to prevent data leakage.
-  **Feature Engineering:** Engineered contextual metrics like Playmaker_Usg, Scoring_Gravity, and Freshman_Star which showed strong predictive power during exploratory data analysis.

3.2 **Methods & Training**

Logistic Regression (Baseline)

Interpretable Baseline

Logistic regression serves as an interpretable model for probabilistic classification. Parameters are learned by minimizing regularized multiclass cross-entropy loss.

Implemented using mini-batch gradient descent with gradient clipping and L2 regularization to prevent overfitting on noisy data.

Probability Formulation

$$p (y = c \mid x) = \frac{\exp (W_c x + b_c)}{\sum_k \exp (W_k x + b_k)}$$

Where $c \in \{0, 1, 2\}$ maps to draft outcomes.

Multilayer Perceptron (MLP)

 **Architecture:** Designed to capture nonlinear feature interactions. Utilizes a single hidden layer of 64 units with ReLU activation.

 **Optimization:** Adam optimizer combined with early stopping (patience of 30 epochs) to strictly prevent overfitting.

 **Imbalance Handling:** Employed sqrt-balanced class weights and evaluated focal loss mechanisms.

Forward Pass Formulation

$$h = \text{ReLU} (W_1 x + b_1) , \quad \hat{p} = \text{softmax} (W_2 h + b_2)$$

K-Nearest Neighbors (KNN)

-  **Similarity Concept:** Classifies players based on historical precedents and statistically similar profiles.
-  **Implementation:** Custom-built using batched NumPy operations and argpartition for optimal neighbor selection.
-  **Tuning:** Optimal parameter found at $k=3$ using Euclidean distance with custom class vote weighting ($w_0=1$, $w_1=w_2=20$).

3.3 Model Evaluation Setup



Data Splits

Time-aware chronological splitting to prevent future data leakage.

Training: 2009-2018. Validation: 2019-2020. Testing: 2021.



Evaluation Metrics

Due to severe class imbalance, accuracy is misleading. Primary metrics are Macro-F1 and Multiclass AUROC. Secondary: Precision, Recall.



Robustness

Threshold tuning on validation set to maximize Macro-F1. Overfitting prevented via L2 regularization, early stopping, and gradient clipping.

3.4 Experimental Results (Test Set)

Metric	Logistic Reg.	MLP (Selected)	KNN (K=3)
Macro-F1	0.4427	0.5306	0.4702
AUROC	0.9416	0.9520	0.7220
Accuracy	0.9471	0.9786	0.9681
Macro-Precision	0.4063	0.4876	0.4280
Macro-Recall	0.6655	0.6346	0.5991

The MLP achieved the highest Macro-F1 and AUROC scores. While Logistic Regression captured higher raw recall (74% of drafted players), the MLP provided a vastly superior balance, dramatically reducing false positives compared to linear models.

3.5 Model Deployment: Streamlit App



Selected Model: MLP deployed for best F1/AUROC balance. Pre-computed weights load instantaneously.



Target Users: NBA scouts, front-office analysts, and coaches looking for data-driven second opinions.

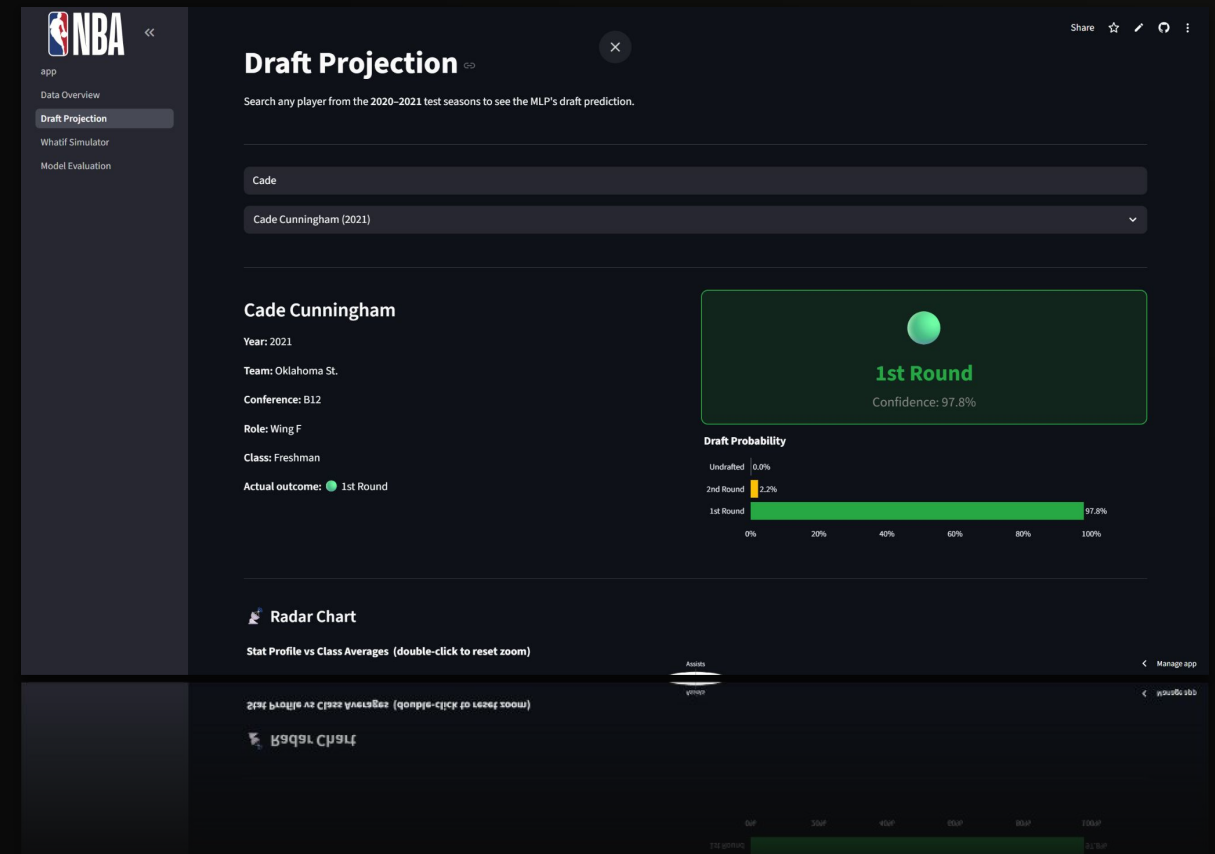


App Interface: 4 core modules: Data Overview, Draft Projection, What-If Simulator, and Model Evaluation.



Performance: Sub-100ms live inference round-trip using our from-scratch NumPy forward pass.

GO TO DEMO



3.6 Conclusion & Future Work



Data Viability

Demonstrated that standard NCAA box scores and engineered metrics contain sufficient signal to meaningfully predict draft categories, democratizing scouting analytics.



Model Insights

Our from-scratch MLP successfully modeled nonlinear interactions to balance precision and recall, whereas Logistic Regression over-predicted and KNN struggled with the imbalance.



Technical Growth

Building models purely via NumPy ensured a deep mathematical understanding of backpropagation, gradient clipping, and custom loss weighting mechanisms.

Future Plans: Incorporate physical combine measurements, expand to 2022+ datasets, and integrate advanced public tracking metrics like RAPTOR or BPM to further elevate recall.